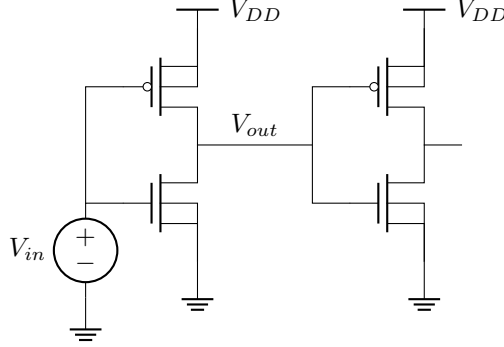


EE5311 Digital IC Design: Assignment 3
CMOS inverter - Transient characteristics and Ring oscillator

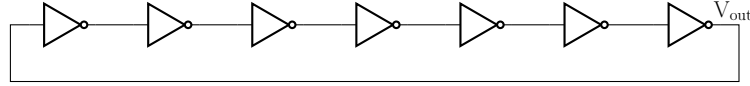
$$V_{Tn} = 0.7\text{V}, \quad \mu_n = 0.025\text{m}^2/\text{V} \cdot \text{s}, \quad C_{oxn} = 8.34\text{fF}/\mu\text{m}^2, \quad v_{satn} = 8 \times 10^4\text{m/s},$$

$$|V_{Tp}| = 0.7\text{V}, \quad \mu_p = 0.009\text{m}^2/\text{V} \cdot \text{s}, \quad C_{oxp} = 8.16\text{fF}/\mu\text{m}^2, \quad v_{satp} = 3 \times 10^4\text{m/s},$$

1. Consider the static CMOS inverter shown below. It drives another identical inverter. The input V_{in} is a pulse between 0 and V_{DD} , with a rise and fall time equal to 5ps and a pulse width of 250ps.



- (a) Set $V_{DD} = 1.8\text{V}$. Assume that $L_n = L_p = 0.15\mu\text{m}$ and $W_n = 0.42\mu\text{m}$. Obtain the delay for $W_p = 0.42\mu\text{m}, 0.84\mu\text{m}, 1.26\mu\text{m}$.
 - (b) From part (a) choose the W_p with minimum delay. Plot the propagation delay(t_p) as a function of V_{DD} for $V_{DD} = 1\text{V}$ to 1.8V in steps of 0.1V . How does it compare with analytical estimates?
 - (c) Plot the measured and estimated energy-delay product as a function of V_{DD} . What is the optimum V_{DD} ?
2. Using the CMOS inverter with the minimum delay from Question 1(a), construct a seven stage ring oscillator as shown below and find the oscillating frequency.



To ensure oscillation in the transient simulation, set the node $V_{out} = 0\text{V}$ initially using: `.ic v(Vout)=0`

- (a) Measure the oscillating frequency for $V_{DD} = 1.8\text{V}$.
- (b) Plot the oscillating frequency and time period as a function of V_{DD} for $V_{DD} = 1\text{V}$ to 1.8V in steps of 0.1V .
- (c) Repeat the experiment (b) with nine inverter stages.